

SCIENCE & TECHNOLOGY

Paper I

Time – 2hrs

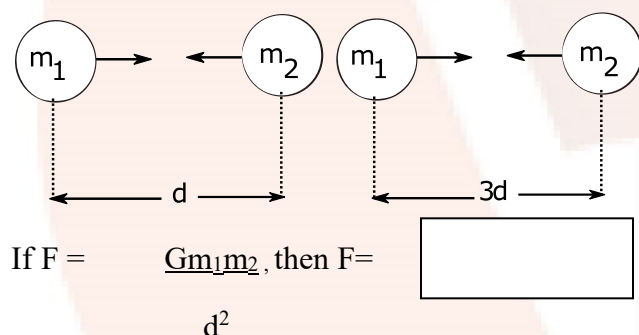
Max. Marks – 40

Note:

1. All questions/activities are compulsory.
2. Figures to the right indicate full marks.
3. Start writing each main question on new page.
4. For each MCQ (i.e Q. No. 1 to 8) evaluation would be done for first attempt only.
5. For each MCQ correct Answer must be written along with its alphabet.
6. Draw scientifically, technically correct labeled diagram wherever necessary.

1. (A) Answer the following questions: [5]

(1) Write the proper Answer in the box:



Answer: Brazil is covered mainly by highlands.

(2) In Dobereiner's triads Li, Na, K the atomic masses of Lithium and potassium are 6.9 and 39.1 respectively then what will be the atomic mass of sodium?

Answer: Atomic mass of Sodium = Mean of the atomic masses of lithium and potassium
 $= (6.9 + 39.1)/2 = 23$

(3) State whether the given statement is true or false:

A concave lens is a converging lens

Answer: False, a concave lens is a diverging lens.

(4) By considering first correlation complete the second correlation:

Hubble Telescope: 569 km high from earth surface

Revolving orbit of Hubble Telescope:

Answer: Revolving orbit of Hubble Telescope: Low earth orbit

(5) Find the odd man out:

Tinning, anodization, alloying, froth flotation.

Answer: Froth flotation because it is a method of extraction of metal from its ore and remaining are the process of prevention of corrosion.

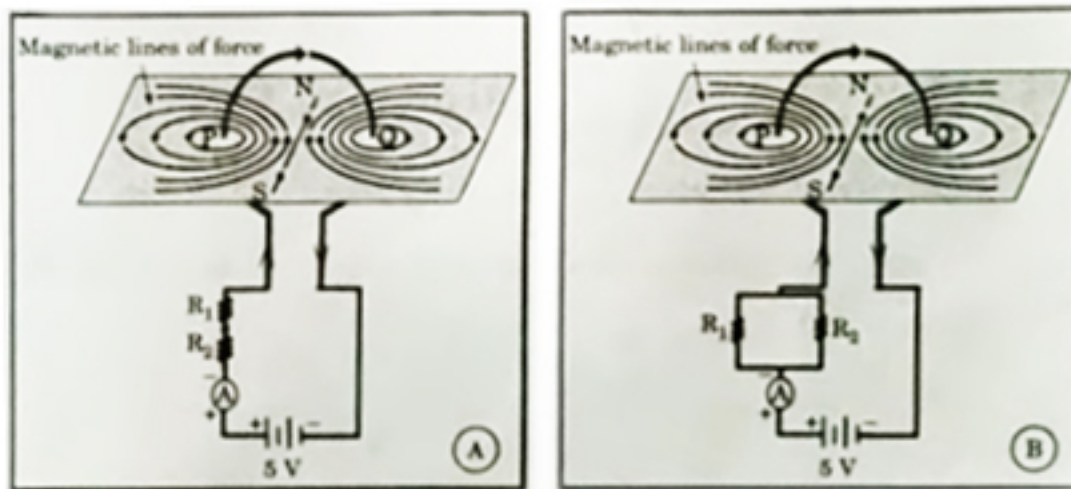
(B) Choose the correct alternative: [5]

(1) The reaction of iron nail with copper sulphate solution is reaction.

- (a) Combination
- (b) Decomposition
- (c) Displacement
- (d) Double Displacement

Answer: (c) Displacement

(2) Observe the following diagram and choose the correct alternative



- (a) The intensity of magnetic field in A is larger than in B
- (b) The intensity of magnetic field in B is less than in A
- (c) The intensity of magnetic field in A and B is same
- (d) The intensity of magnetic field in A is less than in B

Answer: (d) The intensity of magnetic field in A is less than in B

(3) A Ray of light makes an angle of 50° with the surface S_1 of the glass slab. Its angle of incidence will be

- (a) 50°
- (b) 40°
- (c) 140°
- (d) 0°

Answer: (b) 40 degree

(4) Water expands on reducing its temperature below degree.

- (a) 0
- (b) 4
- (c) 8
- (d) 12

Answer: (b) 4. It is called anomalous expansion of water.

(5) The carbon compound used in daily life is

- (a) Edible oil
- (b) Salt
- (c) Carbon dioxide
- (d) Baking soda

Answer: (a) Edible oil

2. Attempt any five of the following questions: [10]

(1) Two tungsten bulbs of power 50 W and 60 W work on 220 volt potential difference. If they are connected in parallel. How much current will flow in the main conductor?

Answer: For 1st tungsten bulb:

Power = 50W

Potential = 220 volt

Power = Potential X Current

Current = Power/Potential = $50/220 = 0.23$

For 2nd tungsten bulb:

Power = 60W

Potential = 220 Volt

Power = Potential X Current

Current = Power/Potential = $60/220 = 0.27$

The flow of current in main conductor = $0.23 + 0.27 = 0.50\text{A}$

(2) Give scientific reason:

In the electric equipment producing heat eg iron, electric heater, boiler, toaster, etc an alloy such as nichrome is used not pure metals

Answer: Alloys such as nichrome have high melting point and resistivity than metals so we use it in the electric equipments.

(3) A metal ball of mass 5 kg falls from a height of 490 meter. How much time it will take to reach the ground? ($g = 9.8 \text{ m/s}^2$)

Answer: Given:

Height (h) = 490m

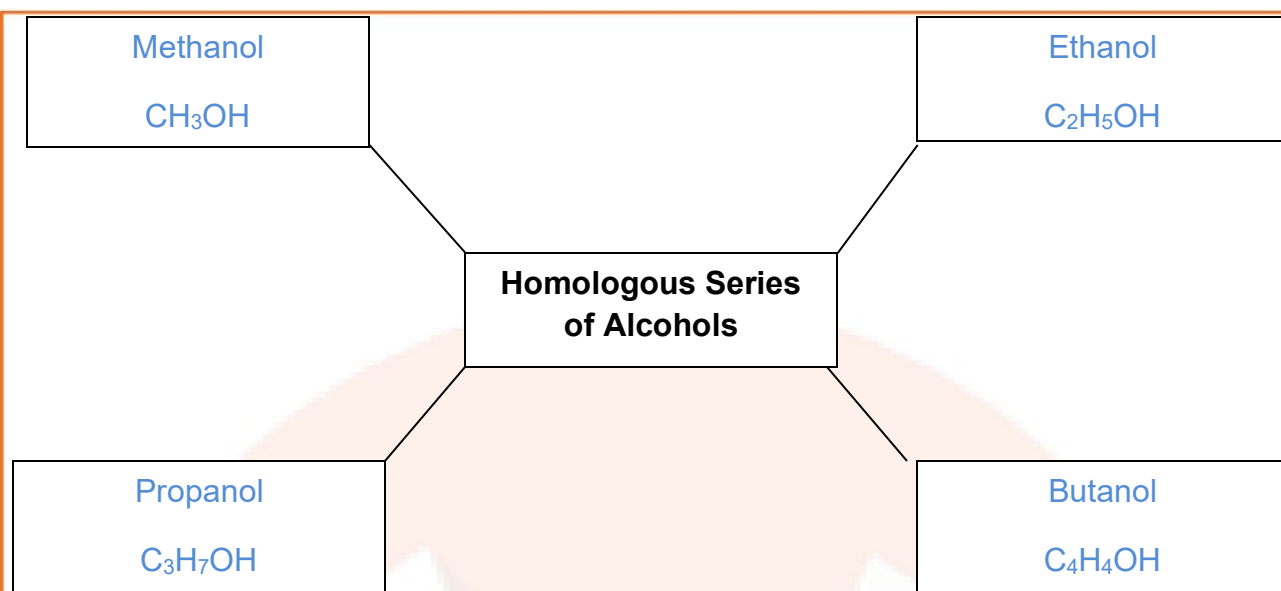
Mass (w) = 5 kg

$g = 9.8\text{m/s}^2$

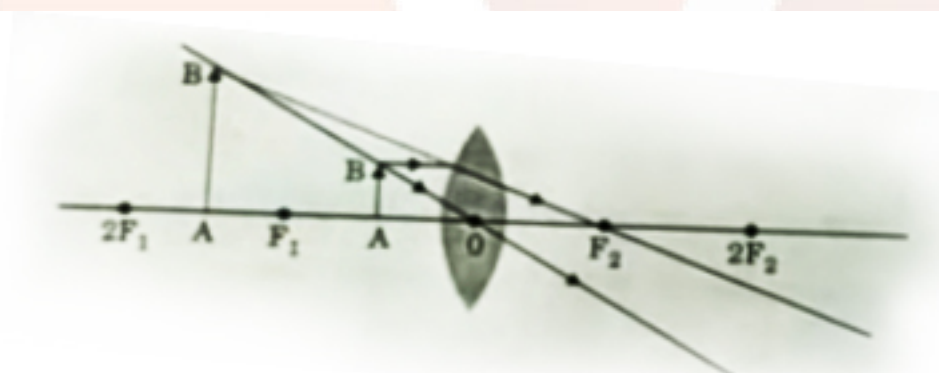
$$\text{Time Taken} = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 490}{9.8}} = \sqrt{100} = 10\text{sec}$$

Therefore, the metal ball will take 10 seconds to reach the ground.

(4) Write names of first four homologous series of alcohols



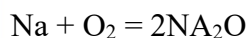
(5) Observe the following figure and complete the table



Position of the object	Between F_1 and O
Position of the image	On the same side of lens as the object
Size of the image	Very large
Nature of image	Vertical and erect

(6) Out of sodium and sulphur, which is a metal explain its reaction with oxygen?

Answer: Sodium is a metal. The reaction of sodium with oxygen is:



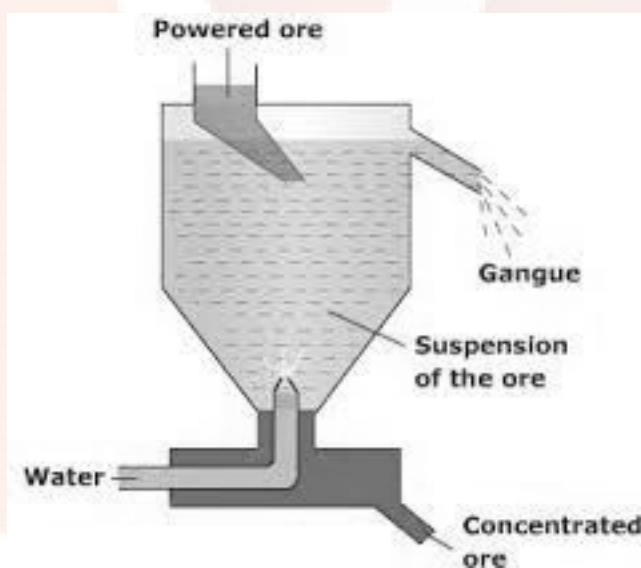
(7) A tapping vessel opens in a tank like container that is tapping on the lower side. The tank has an outlet for water on the upper side and a water inlet on a lower side. Finely ground ore is released in the tank. Forceful jet of water is introduced in the tank from lower side and gangue particles and pure ore are separated by this method.

- The above description is of which gravitational separation method?
- Draw a labeled diagram of this.

Answer:

(a) The above description is of Hydraulic Separation Method.

(b) The diagram of Hydraulic Separation Method is shown below:



3. Attempt any five of the following questions: [15]

- What would be the value of g on the surface of the Earth if its mass was doubled and its radius half what it is now.

Answer: $g = Gm/r^2 \dots\dots\dots(1)$

$$m = 2m \text{ and } r = r/2$$

$$g = G2m/(r/2)^2$$

$$g = G2m/r^2/4$$

$$g = G8m/r^2$$

$$g = 8GM/r^2$$

from (1)

$$g = 8g$$

$$g = 8 \times 9.8 = 78.4 \text{ m/s}^2$$

(2) Write merits of Mendeleev's periodic table.

Answer: The merits of Mendeleev's periodic table are:

- (a) The systematic study of elements as the elements are arranged in order of their increasing atomic masses.
- (b) The table helped in correcting the errors in the atomic masses of certain elements.
- (c) He predicted the properties of those missing elements from the known properties of the other elements in the same group.
- (d) The table also positions the noble gases that were discovered very late.

(3) Study the following chemical reaction and Answerwer the question given below



a. Identify and write the type of chemical reaction.

Answer: Double displacement reaction

b. Write the definition of above type of chemical reaction.

Answer: Double displacement reaction is the chemical reaction in which one component of both of the reacting molecules is exchanged to form the products. During this reaction, the cations and anions of two different compounds change their places, resulting in two entirely different compounds.

c. Write the names of reactants and products of above reaction.

Answer: Reactants:

AgNO_3 – Silver Nitrate

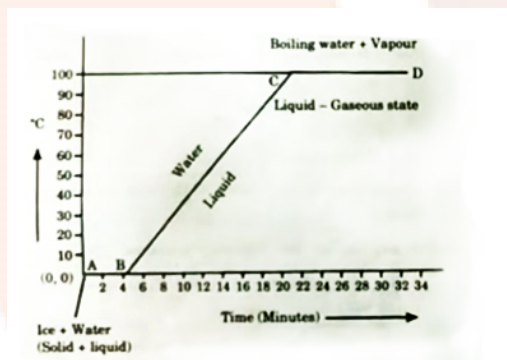
NaCl – Sodium Chloride

Products:

AgCl – Silver Chloride

NaNO_3 – Sodium Nitrate

(4) Explain the following temperature versus time graph



Answer: Temperature vs time graph:

- In this graph, line AB shows conversion of ice into water at constant temperature. So, AB represents: ice + water
- The temperature remains constant during the transition. The transition temperature at which ice changes to water is called melting point of ice.
- Line BC shows the rise in temperature of water from 0° to 100° C. This line represents: solid state $\rightarrow$ liquid state.
- Thereafter, heat energy is supplied to water. The heat energy absorbed by water is used to break the bonds of liquid molecules and convert the liquid into gaseous state.

v. Point C represents: liquid state $\rightarrow$ gaseous state

vi. The constant temperature at which the liquid transforms into gaseous state is called the boiling point of the liquid.

vii. At last, the water completely changes to vapor.

viii. This whole process is called latent heat of vaporization.

(5) Surbhi from standard 10 uses spectacle, the power of the lens in her spectacle is 0.5 D. Answer the following questions from the given information:

i. Identify the type of lenses used in spectacles.

Answer: Convex lenses have been used in her spectacles.

ii. Identify the defect of vision she should be suffering from.

Answer: She is suffering from hypermetropia, far sightedness.

iii. Find the focal length of the lenses.

Answer: The focal length of the lenses = $1/\text{power of lens}$

Where power is 0.5

Therefore, focal length = $1/0.5 = 2$ meters

(6) Complete the following table:

Sr. No	Common Name	Structural Formula	IUPAC Name
1	Ethylene	$\text{CH}_2=\text{CH}_2$	
2		CH_3COOH	Ethanoic Acid
3	Methyl Alcohol		Methanol

Answer:

Sr. No	Common Name	Structural Formula	IUPAC Name
1	Ethylene	$\text{CH}_2=\text{CH}_2$	Ethene
2	Acetic Acid	CH_3COOH	Ethanoic Acid
3	Methyl Alcohol	CH_3OH	Methanol

(7) What is meant by space debris? Why there is need to manage the debris?

Answer: Space debris means the natural debris found in the solar system like asteroids, comets, and meteoroids. The old satellites and spent rocket stages, as well as the fragments from their disintegration and collisions are included in space debris. Space debris is also known as orbital debris, space junk, space waste, space trash, and space litter or space garbage. It is important to manage debris to stop the collision between the existing satellite in space and garbage and between the space craft which is going to moon or other asteroids.

4. Answer any one of the following questions: [5]

(1) Taking into consideration the period of the elements given below, Answer the following questions:

Elements	Atomic Radius (pm)
O	66
B	88
C	77
N	74
Be	111
Li	152

(i) Arrange the shown elements in a decreasing order of their atomic radius.

Answer:

Elements	Atomic Radius (pm)
Li	152
Be	111
B	88
C	77
N	74
O	66

(ii) State the period to which the above element belong.

Answer: Period 2

(iii) Why the arrangement of elements is similar to the above period of modern periodic table?

Answer: All the elements have only two shells in their electronic configuration.

Like

Li (3) – 2, 1

C (6) – 2, 4

N (7) – 2, 5

O (8) – 2, 6

(iv) Which of the above elements have the biggest and the smallest atom?

Answer: Biggest Atom – Li

Smallest Atom – O

(v) What is the periodic trend observed in the variation of atomic radius while going from left to right within a period.

Answer: Atomic radius decreases while going from left to right within a period.

(2) The observations made by Swarali while doing the experiment are given below. Based on these write Answers to the questions.

Swarali found that the light ray travelling from denser medium to rarer medium goes away from the normal if the angle of incidence (i) is raised by Swarali, the angle of refraction (r) went on increasing. However after certain value of the angle of incidence, the light ray is seen to return back into the dense medium.

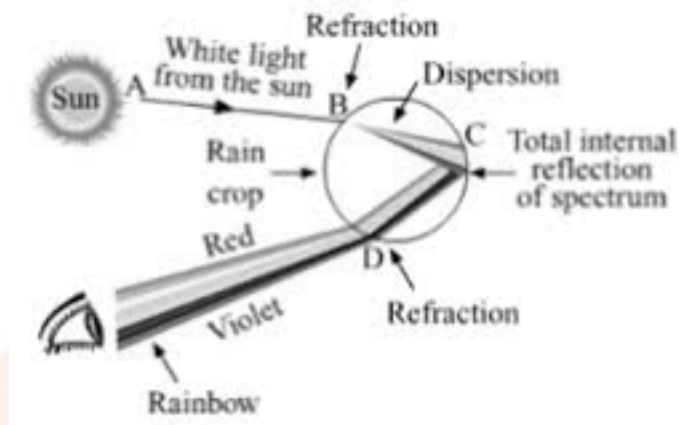
i. What is the specific value of angle i called?

Answer: Critical Angle

ii. What is the process of refraction of incident ray into denser medium called?

Answer: Total Internal Reflection

iii. Draw the diagram of three observations made by Swarali.



SCIENCE AND TECHNOLOGY

PAPER-II

TIME: 2 HOURS

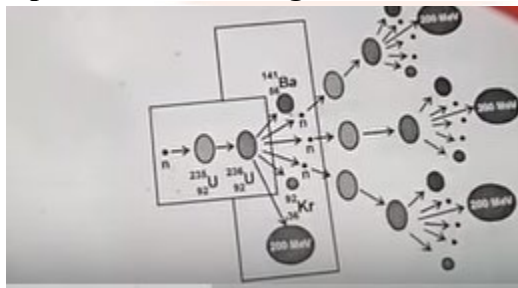
TOTAL MARKS: 40

Note:

- All questions are compulsory.
- Draw scientifically well labelled diagram wherever necessary.
- Start writing each main question on a new page.
- Figure on the right indicate full marks.
- For each MCQ evaluation would be done with one attempt only.
- For Each MCQ correct Answer will be written along with its alphabet
Ex. I (a)..... ii) b).....

1. A) Solve the following questions. [5]

- (i) Identify the process shown in figure and name it. [1]



Answer: It is a nuclear fission reaction.

- (ii) **Pranav and Praniti are twins in your class. They belong to twin type. [1]**

Answer: They belong to dizygotic twin type.

Explanation-Since two different oocytes are fertilized producing two different zygotes .In this case gender can be different .In monozygotic type both gender will be same(i.e. Boy- boy/girl-girl).

- (iii) **There is an oil on the water surface of of river in your area. what will you do?[1]**

Answer: We can clear the oil with the help of microbes such as Pseudomonas species or alcanivorax borkumensis. [1].

- (iv) **Fill in the blank boxes with the help of given clue consumption of alcohol and tobacco material. [1]**

A	D	D	I	C	T	I	O	N
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- (v) **Find out the correlation**

White Revolution: increase in dairy production:

Answer: Green Revolution :Increase in agricultural production.

B) Choose the correct alternative and rewrite the statement: [5]

i) Somatic and stem cells undergo type of division.

- a) Meosis
- b) Mitosis
- c) budding
- d) Cloning

Answer: Somatic and stem cells undergo Mitosis type of division.

ii) Which of the following is a man-made disaster.

- a) Earthquake
- b) Flood

- c) meteoric
- d) Leakage of toxic gases

Answer: Leakage of toxic gases is a manmade disaster.

iii) In a food chain autotrophic plants are present at the..... level.

- a) Nutrition
- b) secondary nutrition
- c) producer
- d) Apex

Answer-In a food chain autotrophic plants are present at the producer level.

iv).....Is a cold blooded animal.

- a) Rat
- b) Snake
- c) Rabbit
- d) Elephant

Answer: Snake Is a cold blooded animal.

v) is a connecting link between Annelida and Arthropoda

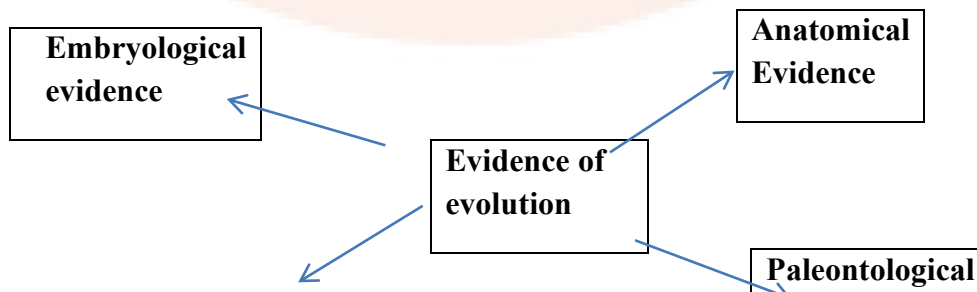
- a) Duck billed platypus
- b) Peripatus
- c) Lungfish
- d) Whale

Answer: Peripatus Is a connecting link between Annelida and Arthropoda.

(vi) Solve any five of the following question.

i) Complete the following chart

Answer:

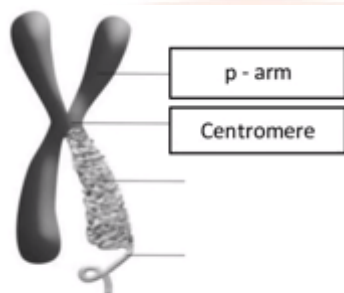


Morphological evidence

Evidence

(ii) Draw any diagram of structure of chromosome and label its parts.

- a) Centromere
- b) P-arm



Structure of Chromosome

(ii) What are advantages of hydroelectric power generation?

Answer: In hydroelectric power generation kinetic energy of flowing water or potential energy of water stored in the water is converted into kinetic energy of water which drives the turbine to produce electrical energy. Following are the advantages of hydroelectric power generation.

- a) There is no pollution is no fuel is burnt during the power generation.
- b) if there is sufficient amount of water in the dam it is possible to generate electricity as per demand.
- c) Water Reservoir can be replenished during rainy season leading to uninterrupted power supply

(iii) State any four benefits of animal classification.

Answer: Following are the four benefits of animal classification.

- a) Study of animal becomes convenient.
- b) We get idea about animal evolution.
- c) Animals can be easily identified with great accuracy.
- d) it helps to understand the relationship with other living organism.
- e) it helps to understand habit of each animal and their ecological role in the nature.

(iv) State any four objectives of disaster management.

Answer: Following are the objectives of disaster management:

- a) Supply of supply of essential commodities to the people to reduce the effect of disaster.
- b) To rehabilitation in disaster victims.
- c) To restore the human life in the region by providing medical treatment.
- d) To take protective and preventive measures before the disaster to reduce its intensity.
- e) To remove the people from affected area.

(v) Give scientific reasons.

Microbial enzymes are used instead of chemical catalyst in a chemical industry.

Answer: Microbial enzymes are used instead of chemical catalyst in chemical industry because

- A) Microbial enzymes are active at low temperature, PH and pressure due to which energy save and erosion free instrumental not necessary.
- b) Enzymes are also carry on specific reactions due to which unnecessary products are not formed due to which it purification cost is minimized.
- C) Enzymes can be reused an eco-friendly

(vi) Read the following extract and Answer the question as follows

Liberal view behind the concept of body and organ donation is that after that death our body should be useful to other needful person so that their miserable life would be comfortable. Awareness about this concept is increasing in our country and people are voluntary vomiting and people are 18 really donating their bodies. Life of many people can be saved by organ and body donation blind can regain division life of many people can be rendered comfortable by donation of organs like liver kidney heart valve, skin etc. Similarly body can be made available for research in medical studies. Many government and social organizations are working toward increasing the awareness about the donation.

(a) What is the liberal view behind the organ and body donation?

Answer: The liberal view behind the organ and body donation is that after that our body should be useful to other needful persons so that their miserable life would become comfortable.

(b) Name any four organs that can be donated.

Answer: Liver, kidney, heart and skin can be donated.

3) Solve the following questions (any five).

[15]

i) Answer the following question.

a) What do you mean by central dogma?

Answer: Information about Protein synthesis which is stored in DNA is transcribed into messenger RNA which further undergoes translation process to produce proteins. This is called as Central dogma.

b) What is transcription?

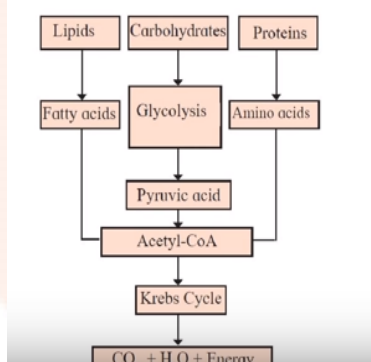
Answer: The process of messenger RNA synthesis from one of the two strands of DNA is called transcription.

c) What is meant by triple codon?

Answer: The three types of nucleotides present in the Messenger RNA which codes for particular amino acid is known as Triple Codon.

ii) Complete the following chart and state which process of energy production it represents.

The given flowchart represents aerobic respiration for the process of energy production.

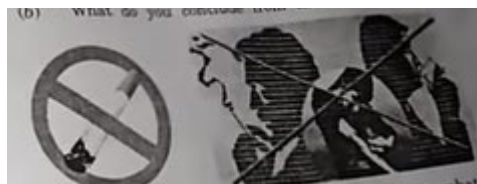


iii)

a) Playing games on mobile while eating is right or wrong justify.

Answer: Playing games on mobile while eating is wrong since it is first done by due to lack of concentration on food in addition to this due to addiction of playing games we can face various physical and mental problems.

b) What do you conclude from the following picture?



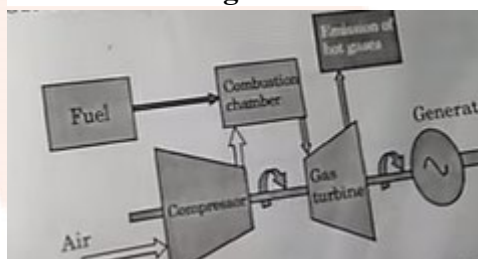
Answer: Cigarette smoking can cause diseases like such as cancer respiratory diseases. Thus we should avoid all the addictive things such as the consumption of alcohol drugs for tobacco related substances which can harm our life.

c) Observe the following picture and state which what can be outcome.



Answer: Selfie on road can lead to accident since people who are addicted to selfies are not aware about world around and risks associated with it.

iv) Observe the diagram and Answerwer the questions.



a) Which energy is generated from the power plant?

Answer: electrical energy generated from the power plant.

b) State its source.

Answer: In this plant the turbine is run by a gas at a very high temperature and pressure generated by combustion of natural gas.

c) Which is more eco-friendly power generation from coal or power generation from natural gas?

Answer: Power generation from natural gas is more eco-friendly since combustion of natural gas leads to less air pollution as compared to gases released during the burning of coal.

v) Identify my class/ phylum and give one example of it.

(a) I have mammary gland and exoskeleton in the form of hair.

Answer: Class Mammalia. Example: Bat, Human Being.

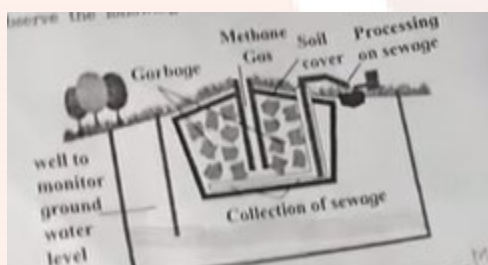
(b) We form the highest number of animals in the planet. We have bilateral symmetry and exoskeleton in the form of chitin.

Answer: Phylum Arthropoda .Example: cockroach crab.

(c) I live in your small intestine my body is large and thread like and pseudocoelomate.

Answer: Phylum Aschelminthes. Example: Ascaris.

(d) Observe the following figure and Answer the question.

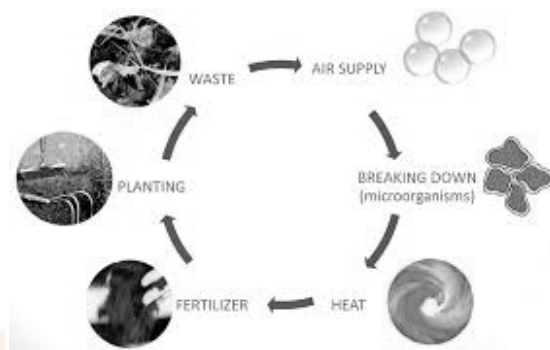


a) Identify the process shown in the figure.

Answer: Modern landfill sites used for compost production process.

b) Explain the process in short.

Answer: The process for compost production is as follows:



(e)

a) What is biotechnology?

Answer: Biotechnology is a branch of science which brings about artificial genetic changes and hybridization in organism for human welfare.

b) Explain any two commercial applications of it.

Answer: Biotechnology in agriculture, biotechnology in human health biotechnology in environment.

Q 4. Solve any one of the following questions

1. a) Identify the following symbols and state their significance.



(i)



(ii)

Answer:

(i) The given symbol represent save water. Water is very essential for living organism thus water saving should be our responsibility for the survival on earth.

ii) The given symbol represents save fuel (Petrol /diesel).Petrol and diesel are nonrenewable source of energy and also burning of petrol and diesel leads to severe air pollution which can harm human health. We should use bicycles instead of using fuel based vehicle.

b) How can biodiversity be conserved.

Answer: Biodiversity diversity can be conserved by following ways:

- i) Protecting the rare species of organism
- ii) Establishing national parks and sanctuaries.
- iii) Observing rules and regulations
- iv) Maintaining record of traditional knowledge.
- v) Declaring some regions as bio reserves.
- vi) Initiative projects from the conservation of plants and animals.

ii) Answer the following questions.

a) Gender of child is determined by the male partner. Draw diagram explaining the above statement. [2]

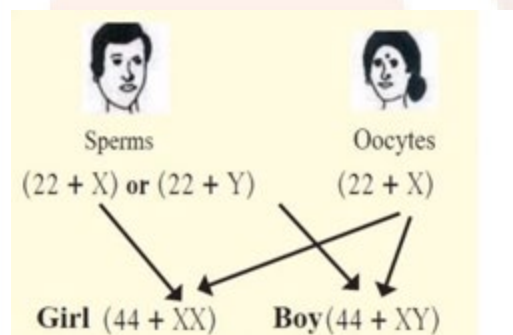
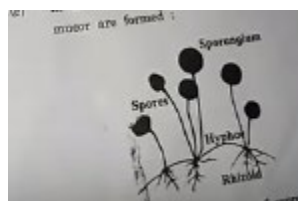


DIAGRAM FOR SEX DETERMINATION IN HUMAN BEING

b. Prepare a slogan for campaign against female foeticide. [1]

Answer: “Save child save girl child “or “stop female foeticide”.

b) In the following figure explain how fungal colonies of new platform [1]



Answer: The spore from sporangia of the fungi is released which then germinate in moist and warm places giving rise to new colony.

c) Identify and state the type of reproduction represented in the above figure. [1]

Answer: Spore formation one type of asexual reproduction in multicellular organism is represented in the figure.

